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Impedance mismatch suppressing connector

Abstract

Provided is a technical system by which the impedance mismatch of the plug unit of a connector can be suppressed. A plug unit of a connector includes: a plurality of contacts; and a housing having a base portion provided with a plurality of through holes into which the plurality of contacts are inserted, and a header that protrudes from the base portion and is provided with a plurality of slits holding portions of the plurality of contacts exposed toward the header side from the through holes. A portion of the contact inserted into the through hole and a portion of the contact held by the slit are formed in a linear shape.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

(1) This patent application claims priority from PCT Patent Application No. PCT/JP2020/032910 filed Aug. 31, 2020. This patent application is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a connector with a floating mechanism.

BACKGROUND

(3) Among the connectors respectively mounted on the circuit board and the expansion board to mediate signal transmission between the boards, some have a floating mechanism that absorbs the error between the attachment position of the connector on the circuit board side and the attachment position of the connector on the expansion board side. As a document disclosing a technique related to this type of connector, Patent Document 1 can be taken up.

(4) A connector for substrate connection of Patent Document 1 is composed of a socket-type first connector and a plug-type second connector fitted into the first connector. The first connector of this Document 1 includes a fixed housing, a floating housing, and a plurality of contacts bridged between these housings. The contact of the first connector has a first spring portion and a second spring portion extending in an inverted V-shape, and a curved inverted portion interposed between their upper ends, and a partition member made of an electrically insulating synthetic resin material is arranged between the inverted curved portions of adjacent contacts. According to the technique of Patent Document 1, due to the action of the partition member arranged between the inverted curved portions, the difference in impedance between the inverted curved portion and the other portion extending linearly is narrowed, and the impedance of the entire contact is homogenized. Thus, according to the Patent Document, the impedance mismatch can be suppressed

PRIOR ART REFERENCE

Patent Document

(5) Patent Document 1: Japanese Patent Application Publication No. 2013-045739

SUMMARY

Problems to be Solved

(6) However, the technique of Patent Document 1 has a mechanism that suppresses impedance mismatch on the socket type connector side, but it does not have such a mechanism on the plug type connector side. For this reason, the impedance mismatch on the plug type connector cannot be suppressed.

(7) The present invention has been made in view of such a problem, and one of the objects is to provide a technical means capable of suppressing the impedance mismatch of the plug unit of a connector.

Means for Solving the Problem

(8) In order to solve the above problem, a preferred aspect of the present disclosure is a connector including: a receptacle unit mounted on a first substrate and a plug unit mounted on a second substrate and connected to the receptacle unit, and electrically connecting the first substrate and the second substrate by connecting the receptacle unit and the plug unit, wherein the plug unit includes: a plurality of contacts; and a housing including a base portion provided with a plurality of through holes into which the a plurality of contacts are inserted, and a header protruding from the base portion and provided with a plurality of slits that hold portions of the a plurality of contacts exposed toward the header side from the through holes, and a portion of the contact inserted into the through hole and a portion of the contact held by the slit are formed in a linear shape.

(9) In this aspect, two holding portions that hold a linear portion of the contact may be provided at an upper end and a lower end of the through hole in the base portion, and a gap may be formed between a peripheral wall of a portion between the two holding portions in the through hole and the linear portion.

(10) Further, the contact has: a linear portion extending linearly along the through hole and the slit;

and a terminal portion bent and extending outward from one end of the linear portion and connected to the substrate.

(11) Further, the plurality of contacts may be in two columns of a left side column and a right side column.

(12) Further, the through hole may have four peripheral walls opposed to each other in a left-right direction and a front-rear direction, and a groove may be provided at a portion of the peripheral wall excluding the two holding portions.

(13) Further, the slit may have a peripheral wall connected to the peripheral wall of the through hole, and a groove may be provided in the peripheral wall of the slit.

(14) Further, the groove of the through hole and the groove of the slit are recessed on the same side.

(15) Further, the groove of the through hole and the groove of the slit have the same depth.

Effects

(16) According to the present invention, a connector includes: a receptacle unit mounted on a first substrate and a plug unit that is mounted on a second substrate and connected to the receptacle unit, and the connector electrically connects the first substrate and the second substrate by connecting the receptacle unit and the plug unit, the plug unit includes: a plurality of contacts; and a housing including a base portion provided with a plurality of through holes into which the plurality of contacts are inserted, and a header protruding from the base portion and provided with a plurality of slits that hold portions of the a plurality of contacts exposed toward the header side from the through holes, and a portion of the contact inserted into the through hole and a portion of the contact held by the slit are formed in a linear shape. Thus, variations in impedance between the linear portion of the contact and the other bent portions are reduced. Therefore, it is possible to suppress impedance mismatch of the connector.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A is a perspective view of a receptacle unit **93** of a connector **100** according to one embodiment of the present disclosure, and FIG. 1B is a perspective view of a plug unit **97** of the connector **100**;

(2) FIG. 2 is a perspective view of the receptacle unit **93** of FIG. 1A viewed from an opposite side;

(3) FIG. 3 is a cross-sectional view of the receptacle unit **93** and the plug unit **97** fitted together;

(4) FIGS. 4A and 4B are diagrams showing how the receptacle unit **93** floats in an X direction, and FIG. 4 (C) is a diagram showing how the receptacle unit **93** floats in a Y direction;

(5) FIG. 5A is a view of the receptacle unit **93** viewed from a +Z side, FIG. 5B is a view of the receptacle unit **93** viewed from a +Y side, FIG. 5C is a view of the receptacle unit **93** viewed from a -Z side, and FIG. 5D is a view of the receptacle unit **93** viewed from an -X side;

(6) FIG. 6 is a cross-sectional view taken along line A-A of FIG. 5B;

(7) FIG. 7A is a perspective view of a fixed housing **1** of the receptacle unit **93**, and FIG. 7B is a cross-sectional view taken along line A-A of FIG. 7A;

(8) FIG. 8A is a perspective view of a floating housing **2** of the receptacle unit **93**, and FIG. 8B is a cross-sectional view taken along line A-A of FIG. 8A;

(9) FIG. 9 is a perspective view of a contact **3-j** of the receptacle unit **93**;

(10) FIG. 10 is a perspective view of an anchor plate **4** of the receptacle unit **93**;

(11) FIG. 11 is a perspective view of a conductive resin member **5** of the receptacle unit **93**;

(12) FIG. 12A is a view of the plug unit **97** viewed from the +Z side, FIG. 12B is a view of the plug unit **97** viewed from the +Y side, FIG. 12C is a view of the plug unit **97** viewed from the -Z side, and FIG. 12D is a view of the plug unit **97** viewed from the -X side;

(13) FIG. 13A is a cross-sectional view taken along line D-D of FIG. 12B, and FIG. 13B is an

enlarged view within the frame G of FIG. 13A;

(14) FIG. 14 is an enlarged view within the F frame of FIG. 12B;

(15) FIG. 15A is a cross-sectional view taken along line H-H of FIG. 12A, and FIG. 15B is an enlarged view within the frame J of FIG. 15A;

(16) FIG. 16 is an enlarged view within the frame K of FIG. 12C;

(17) FIG. 17A is a perspective view of a housing 6 of the plug unit 97, and FIG. 17B is a cross-sectional view taken along line A-A of FIG. 17A;

(18) FIG. 18 is a perspective view of a contact 7-j of the plug unit 97; and

(19) FIG. 19 is a perspective view of an anchor plate 8 of the plug unit 97.

DETAILED DESCRIPTION

(20) Hereinafter, a connector 100 according to one embodiment of the present disclosure is explained with reference to drawings. The connector 100 includes a receptacle unit 93 and a plug unit 97. The receptacle unit 93 is mounted on an electronic substrate, which is a first substrate. The plug unit 97 is mounted on an expansion board, which is a second substrate. By fitting a header 62 of the plug unit 97 into a slot 20 of the receptacle unit 93 and connecting the receptacle unit 93 and the plug unit 97, K (K is 144, for example) contacts 7-j (j=1 to K) of the plug unit 97 and K contacts 3-j (j=1 to K) of the receptacle unit 93 are electrically connected, respectively, and high-speed differential transmission of maximum number of Gbps due to PAM (Pulse Amplitude Modulation) is made possible.

(21) In the following description, a fitting direction of the plug unit 97 with respect to the receptacle unit 93 is appropriately referred to as a Z direction, one direction orthogonal to the Z direction is appropriately referred to as an X direction, and a direction orthogonal to both the Z direction and the Y direction is appropriately referred to as a Y direction. In addition, the +Z side may be referred to as an upper side, the -Z side may be referred to as a lower side, the +X side may be referred to as a front side, the -X side may be referred to as a rear side, the +Y side may be referred to as a left side, and the -Y side may be referred to as a right side. Further, among the contacts 3-j (j=1 to K) of the receptacle unit 93 and the contacts 7-j (j=1 to K) of the plug unit 97, letter (G) is attached to the contacts 3-j and contacts 7-j for ground of differential transmission, and letter (S) is attached to the contacts 3-j and contacts 7-j for signal to distinguish between them.

(22) As shown in FIG. 1A, FIG. 2, FIG. 5A, FIG. 5B, FIG. 5C, and FIG. 5D, the receptacle unit 93 has a fixed housing 1, a floating housing 2, contacts 3-j (j=1 to K), an anchor plate 4, and a conductive resin member 5.

(23) As shown in FIG. 7, the fixed housing 1 has an opening portion 10 penetrating through the fixed housing 1 upward and downward, and wall portions 11 and 12, which are first wall portions surrounding the opening portion 10. The wall portions 11 are opposed to each other in the X direction with the opening portion 10 sandwiched therebetween, and the wall portions 12 are opposed to each other in the Y direction with the opening portion 10 sandwiched therebetween.

(24) The upper and left and right edges of the outer surfaces of the front and rear wall portions 11 of the fixed housing 1 protrude outward as convex portions 111. In the middle of the outer surface of the wall portion 11, there is a middle plate 112 extending upward and downward. A positioning boss 113 is provided at the center of the lower surface of the wall portion 11. This positioning boss 113 is fitted into a positioning hole of an electronic substrate when fixing the receptacle unit 93 to the electronic substrate.

(25) Holders 121 for mounting the anchor plate 4 are provided at end portions on the front side and rear side of the outer surface of the left and right wall portions 12 of the fixed housing 1. A lower end side between the two holders 121 on the outer surface of the wall portion 12 protrudes outward as a convex portion 122.

(26) The lower sides of the inner surfaces of the left and right wall portions 12 of the fixed housing 1 are respectively provided with a column of K/2 slits 17-j arranged in the front and rear. On the upper sides of the inner surfaces of the wall portions 12 on the left and right, there are curved

portions **14** curved inward in a quarter arc-shape, and there are inner edge portions **15** parallel to the inner surfaces on the curved portions **14**.

(27) As shown in FIG. **8A** and FIG. **8B**, the floating housing **2** has a slot **20** into which the header **62** of the floating housing **2** is inserted, and wall portions **21** and **22**, which are the second wall portions constituting the inner circumferential surface of the slot **20**. The wall portions **21** are opposed to each other in the X direction with the slot **20** sandwiched therebetween, and the wall portions **22** are opposed to each other in the Y direction with the slot **20** sandwiched therebetween. The floating housing **2** has dimensions to be accommodated in the opening portion **10** of the fixed housing **1**.

(28) The upper and left and right edges of the outer surfaces of the front and rear wall portions **21** of the floating housing **2** protrude outward as convex portions **211**. The left and right lower ends of the convex portion **211** protrude outward.

(29) The inner surfaces of the left and right wall portions **22** of the floating housing **2** are stepped surfaces in which the surface on the lower side protrudes with respect to the surface on the upper side. The lower sides of the inner surfaces of the left and right wall portions **22** of the floating housing **2** are respectively provided with a column of K/2 slits **26-j** arranged in the front and rear. On the upper sides of the outer surfaces of the left and right wall portions **22**, there are curved portions **24** curved outward in a quarter arc-shape, and there are outer edge portions **25** parallel to the outer surfaces on the curved portions **24**.

(30) As shown in FIG. **5A**, FIG. **5C**, and FIG. **8B**, a bottom portion **23** forming the bottom of the slot **20** is bridged between the lower end portions of the front and rear wall portions **21** in the floating housing **2**. The bottom portion **23** extends in a straight line in the front and rear. As shown in FIG. **8B**, there are through holes **220** between the bottom portion **23** and the inner surfaces of the wall portions **22** on both left and right side thereof. The through holes **220** penetrate through from the bottom surface toward the inside of the slot **20**. The bottom surface of the floating housing **2** is provided with a conductive resin member **5**. The bottom surface of the bottom portion **23** of the floating housing **2** is provided with a groove **230** for fixing the conductive resin member **5**.

(31) The contacts **3-j** ($j=1$ to K) form a column of K/2 contacts **3-j** on the left and a column of K/2 contacts **3-j** on the right. As shown in FIG. **9**, the contact **3-j** has a terminal portion **31** extending in a left-right direction on one end side, a contact point portion **32** extending in an up-down direction on the other end side, a support piece portion **36** connected to the terminal portion **31**, a support piece portion **37** connected to the contact point portion **32**, and an elastic deformation portion **38** interposed between these two support piece portions **36** and **37**.

(32) The tip end of the contact point portion **32** is bent into a doglegged shape. There are claw portions **361** and **371** protruding outward on the side surfaces of the support piece portion **36** and the support piece portion **37**.

(33) The elastic deformation portion **38** has a flexible portion **380** rounded in a semicircular shape, two linear portion **381** and linear portion **382** extending downward from both ends of the flexible portion **380**, a connecting portion **386** bent toward the support piece portion **36** side at the lower end of the linear portion **381** and connected to the upper end of the support piece portion **36**, and a connecting portion **387** bent toward the support piece portion **37** side at the lower end of the linear portion **382** and connected to the lower edge of the support piece portion **37**. The elastic deformation portion **38** can be elastically deformed frontward, rearward, leftward, and rightward.

(34) As shown in FIG. **10**, the anchor plate **4** has a main body portion **41**, and a protruding portion **42** protruding downward from the lower edge of the main body portion **41**. The upper edge of the side surface of main body portion **41** is provided with a convex portion **411** expanding laterally. There are claw portions **412** protruding outward below the convex portion **411** on the side surface of the main body portion **41**. The protruding portion **42** is provided with a long hole **421**.

(35) As shown in FIG. **11**, the conductive resin member **5** has a lower plate portion **51** in which four corners of a rectangular parallelepiped shape are protruded outward, a convex portion **53**

protruding upward from the center of the upper surface of the lower plate portion **51**, and pillar portions **59** erecting upward from tip ends of the lower plate portion **51** that are protruded outward. The left and right width of the projection **53** is substantially the same as the left and right width of the groove **230** of the bottom portion **23** of the floating housing **2**.

(36) The fixed housing **1**, the floating housing **2**, the contacts **3-j** ($j=1$ to K), the anchor plate **4**, and the conductive resin member **5** are combined as follows.

(37) As shown in FIG. 5A, FIG. 5C, and FIG. 6, the floating housing **2** is accommodated in the opening portion **10** of the fixed housing **1**, two columns of contacts **3-j** on the left and right are bridged between the floating housing **2** and the fixed housing **1**, the conductive resin member **5** is mounted in the groove **230** of the bottom portion **23** of the floating housing **2**, and the anchor plate **4** is mounted on the holder **121** of the fixed housing **1**.

(38) Here, in FIG. 5C, in order to show the groove **230**, the number of the conductive resin members **5** is six, but in reality, the conductive resin members **5** are mounted on the entire lower surface of the bottom portion **23**.

(39) As shown in FIG. 5A, between the wall portion **11** of the fixed housing **1** and the wall portion **21** of the floating housing **2** in the opening portion **10**, there is a movable space **101** for making the floating housing **2** float forward and rearward, and between the wall portion **12** of the fixed housing **1** and the wall portion **22** of the floating housing **2**, there is a movable space **102** for making the floating housing **2** float leftward and rightward. The contacts **3-j** ($j=1$ to K) are arranged in such an arrangement order that two contacts **3-j** (G) and two contacts **3-j** (S) are taken as one group, and the two contacts for signal are accommodated between the two contacts for ground.

(40) As shown in FIG. 6, the contact **3-j** is press-fitted into the fixed housing **1** and the floating housing **2** from the lower side such that the contact point portion **32** is protruded into the slot **20** via the through hole **220**, and the elastic deformation portion **38** is accommodated in the movable space **102**. The support piece portion **36** of the contact **3-j** is accommodated and supported in the slit **17-j**. The support piece portion **37** of the contact **3-j** is accommodated and supported in the slit **26-j**.

(41) The connecting portion **387** of the contact **3-j** is in contact with the lower surface of the floating housing **2**. The terminal portion **31** of the contact **3-j** is exposed to the lower side of the lower surface of the fixed housing **1**. The terminal portion **31** of the contact **3-j** is soldered and fixed to the pad of the electronic substrate. Between the adjacent contacts **3-j**, there is no partition of the electric insulating member, and only an air layer exists.

(42) As shown in FIG. 6, a gap **120** is provided between the inner edge portion **15** of the fixed housing **1** and the outer edge portion **25** of the floating housing **2**. On the lower side of the gap **120** in the movable space **102**, the curved portion **14** of the wall portion **12** and the curved portion **24** of the wall portion **22** are curved along the shape of the flexible portion **380** of the elastic deformation portion **38**.

(43) The convex portion **53** of the conductive resin member **5** is fitted in the groove **230** of the bottom portion **23** of the floating housing **2**. The two pillar portions **59** on the left side of the conductive resin member **5** are in contact with the two contacts **3-j** (G) on the left side. The two pillar portions **59** on the right side of the conductive resin member **5** are in contact with the two contacts **3-j** (G) on the right side. The four contacts **3-j** (G) are electrically connected via the conductive resin member **5**. The protruding portion **42** of the anchor plate **4** mounted on the holder **121** extends to the lower side of the lower surface of the fixed housing **1**.

(44) As shown in FIG. 1B, FIG. 12A, FIG. 12B, FIG. 12C, and FIG. 12D, the plug unit **97** has a housing **6**, contacts **7-j** ($j=1$ to K), and an anchor plate **8**.

(45) As shown in FIG. 17A and FIG. 17B, the housing **6** has a header **62** and a base portion **63**.

(46) The header **62** protrudes downward from the lower surface of the base portion **63**. The left and right outer surfaces of the header **62** are each provided with a column of $K/2$ slits **66-j**.

(47) The slit **66-j** on the left side has a U-shaped cross section surrounded by the right peripheral wall, the front peripheral wall, and the rear peripheral wall. The right side of the slit **66-j** on the left

side is open, and the front peripheral wall and the rear peripheral wall are opposed to each other. The slit **66-j** on the right side has a U-shaped cross section surrounded by the left peripheral wall, the front peripheral wall, and the rear peripheral wall. The left side of the slit **66-j** on the right side is open, and the front peripheral wall and the rear peripheral wall are opposed to each other.

(48) As shown in FIG. **17B**, grooves **661** are provided in three peripheral walls of the slits **66-j** on the left side and right side. The groove **661** of the front peripheral wall is recessed forward, the groove **661** of the rear peripheral wall is recessed rearward, the groove **661** of the left peripheral wall is recessed leftward, and the groove **661** of the right peripheral wall is recessed rightward.

(49) A positioning boss **613** is provided at the center of the front and rear end portions of the upper surface of the base portion **63**. The positioning boss **613** is fitted into the positioning hole of the expansion board when fixing the plug unit **97** to the expansion board. The upper edges of the front and rear side surfaces of the base portion **63** expand outward as outer edge portions **631**. The outer edge portion **631** is provided with a holder **632** for mounting the anchor plate **8**.

(50) A column of $K/2$ through holes **67-j** arranged in the front and rear is provided on each of the left side and right side of the center of the base portion **63** in the Y direction. The through hole **67-j** penetrates through the base portion **63** upward and downward. The through hole **67-j** on the left side is communicated with the slit **66-j** on the left side, and the through hole **67-j** on the right side is communicated with the slit **66-j** on the right side.

(51) The through holes **67-j** on the left side and right side have a rectangular cross section surrounded by the left peripheral wall, the right peripheral wall, the front peripheral wall, and the rear peripheral wall. The left peripheral wall and the right peripheral wall are opposed to each other, and the front peripheral wall and the rear peripheral wall are opposed to each other, in the through holes **67-j** on the left side and right side. The peripheral wall of the through hole **67-j** is connected to the peripheral wall of the slit **66-j**.

(52) As shown in FIG. **15A**, two holding portions **679** and **678** constricted inward are provided at the upper end and lower end of the through holes **67-j** on the left side and right side.

(53) As shown in FIG. **17B**, grooves **672** are provided in portions of the four peripheral walls of the left and right through holes **67-j** excluding the two holding portions **679** and **678**. The groove **672** of the front peripheral wall is recessed forward, the groove **672** of the rear peripheral wall is recessed rearward, the groove **672** of the left peripheral wall is recessed leftward, and the groove **672** of the right peripheral wall is recessed rightward. The orientation and depth of the groove **672** of the through hole **67-j** are the same as the orientation and depth of the groove **661** of the slit **66-j**.

(54) The contacts **7-j** ($j=1$ to K) form a column of $K/2$ contacts **7-j** on the left and a column of $K/2$ contacts **7-j** on the right. As shown in FIG. **18**, the contact **7-j** has a linear portion **72** extending in the up-down direction on one end side, a terminal portion **79** extending in the left-right direction on the other end side, and a connecting portion **80** interposed between the linear portion **72** and the terminal portion **79**.

(55) There is a claw portion **720** protruding outward at a position away from the lower end toward the upper side on the side surface of the linear portion **72**. The upper end of the linear portion **72** is slightly wider than the portion below it. Further, this wider portion is provided with a long hole **721**.

(56) As shown in FIG. **19**, the anchor plate **8** has two rectangular piece portions **81**, a bridged portion **40** bridged between the lower ends of the two rectangular piece portions **81**, and protruding portions **83** protruding upward from the upper edges of the two rectangular piece portions **81**. There are claw portions **812** protruding outward on the side surfaces of the two rectangular piece portions **81**.

(57) The housing **6**, the contacts **7-j** ($j=1$ to K), and the anchor plate **8** are combined as follows. The contacts **7-j** ($j=1$ to K) are press-fitted into the housing **6** from the upper side. The linear portion **72** of the contact **7-j** is inserted into the through hole **67-j**, and its tip end side passes through the through hole **67-j** and is held in the slit **66-j**. The terminal portion **79** of the contact **7-j** is exposed

on the upper side of the upper surface of the housing **6**. The terminal portion **79** of the contact **7-j** is soldered and fixed to the pad of the expansion board.

(58) The anchor plate **8** is mounted on the holder **632** of the housing **6**. The protruding portion **83** of the anchor plate **8** extends to the upper side of the upper end of the housing **6**. The linear portion **72** of the contact **7-j** extends in a straight line along the through hole **67-j** and the slit **66-j**. The portion of the linear portion **72** where the claw portion **720** is located is supported by the holding portion **678** of the lower side, and the thicker upper end portion of the linear portion **72** is supported by the holding portion **679** of the upper side.

(59) As shown in FIG. **3**, when the header **62** of the plug unit **97** and the slot **20** of the receptacle unit **93** are fitted together, the linear portion **72** of the contact **7-j** of the plug unit **97** and the contact point portion **32** of the contact **3-j** of the receptacle unit **93** come into contact with each other.

(60) As shown in FIG. **4A** and FIG. **4B**, when the relative positions of the plug unit **97** and the receptacle unit **93** are displaced from each other in the front-rear direction, the contact **3-j** is twisted so that the linear portion **381** and the linear portion **382** are tilted to the opposite sides, and the receptacle unit **93** moves forward or backward in the opening portion **10**. Thereby, the positional displacement in the front-rear direction is eliminated.

(61) Further, as shown in FIG. **4C**, when the relative positions of the plug unit **97** and the receptacle unit **93** are displaced from each other in the left-right direction, the left and right contacts **3-j** are twisted so that one of the left and right linear portion **381** and linear portion **382** is widened and the other one of the linear portion **381** and linear portion **382** is narrowed, and the receptacle unit **93** moves leftward or rightward in the opening portion **10**.

(62) The details of the configuration of the present embodiment are described above. The plug unit **97** of the connector **100** according to the present embodiment includes: a plurality of contacts **7-j** ($j=1$ to K); and a housing **6** having a base portion **63** provided with a plurality of through holes **67-j** ($j=1$ to K) into which the plurality of contacts **7-j** ($j=1$ to K) are inserted, and a header **62** protruding from the base portion **63** and provided with a plurality of slits **66-j** ($j=1$ to K) that hold portions of the plurality of contacts **7-j** ($j=1$ to K) exposed toward the header **62** side from the through holes **67-j** ($j=1$ to K). A portion of the contact **7-j** inserted into the through hole **67-j** and a portion of the contact **7-j** held by the slit **66-j** are formed in a linear shape. Two holding portions **679** and **678** that hold a linear portion **72** of the contact **7-j** are provided at an upper end and a lower end of the through hole **67-j** in the base portion **63**, and a gap is formed between a peripheral wall of a portion between the two holding portions **679** and **678** in the through hole **67-j** and the linear portion **72** by a groove **672** between the peripheral wall and the linear portion **72**. Thus, variations in impedance between the linear portion **72** of the contact **7-j** and the other bent portions are reduced. Therefore, it is possible to suppress impedance mismatch of the plug unit **97**.

(63) Although the embodiment of the present disclosure has been described above, the following modifications may be added to this embodiment.

(64) (1) In the embodiment described above, the column of the contacts **7-j** ($j=1$ to K) of the plug unit **97** may be 1 column, or may be three columns or more.

(65) (2) In the embodiment described above, the number of the anchor plates **8** is not necessary to be two, and may be one, or may be three or more.

(66) (3) In the embodiment described above, grooves **672** are provided in portions of the four peripheral walls of the left and right through holes **67-j** excluding the two holding portions **679** and **678**. However, it is not necessary to provide the grooves **672** in all the four peripheral walls of the left and right through holes **67-j**. For example, the grooves **672** may be provided only in the two left and right peripheral walls of the left and right through holes **67-j**, or may be provided only in the two front and rear peripheral walls.

(67) (4) In the embodiment described above, grooves **661** are provided in three peripheral walls of the left and right slits **66-j**. However, it is not necessary to provide the slits **661** in all the three peripheral walls of the left and right through holes **66-j**. For example, the grooves **661** may be

provided only in the peripheral walls of the left and right slits 66-j on the side opposite to the open side.

(68) (5) In the embodiment described above, the number K of the contacts 7-j may be 145 or more, or may be 143 or less.

Claims

1. A connector comprising a receptacle unit mounted on a first substrate and a plug unit that is mounted on a second substrate and is connected to the receptacle unit, wherein, the receptacle unit comprises a plurality of first contacts; the plug unit comprises a plurality of second contacts; the connector electrically connects the first substrate and the second substrate by connecting the receptacle unit and the plug unit via the plurality of first contacts and the plurality of second contacts; and the plug unit further comprises; a housing comprising a base portion provided with a plurality of through holes into which the plurality of second contacts are inserted, and a header that protrudes from the base portion and is provided with a plurality of slits to hold portions of the second contacts, the portions being exposed toward the header side from the through holes, and wherein, a first portion of the second contact inserted into the through hole and a second portion of the second contact held by the slit are formed in a linear shape.
 2. The connector according to claim 1, wherein two holding portions that hold a linear portion of the second contact are provided at an upper end and a lower end of the through hole in the base portion, and a gap is formed between the linear portion and a peripheral wall of a portion between the two holding portions in the through hole.
 3. The connector according to claim 1, wherein the second contact comprises: a linear portion extending linearly along the through hole and the slit; and a terminal portion that is bent and extending outward from one end of the linear portion and is connected to the substrate.
 4. The connector according to claim 3, wherein the plurality of second contacts are in two columns of a left side column and a right side column.
 5. The connector according to claim 4, wherein the through hole comprises four peripheral walls opposed to each other in a left-right direction and a front-rear direction, and a groove is provided at a portion of the peripheral wall excluding the two holding portions.
 6. The connector according to claim 5, wherein the slit comprises a peripheral wall connected to the peripheral wall of the through hole, and a groove is provided in the peripheral wall of the slit.
 7. The connector according to claim 6, wherein the groove of the through hole and the groove of the slit are recessed on the same side.
 8. The connector according to claim 7, wherein the groove of the through hole and the groove of the slit have the same depth.
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